



RV College of Engineering[®]

Mysore Road, RV Vidyaniketan Post, Bengaluru - 560059, Karnataka, India

**Self-Reflective Agentic AI: Enhancing Autonomous Decision-Making through Internal
Self-Evaluation**

TECHNICAL SEMINAR REPORT

MCA492L

submitted by

Swasthik Devadiga 1RV24MC108

under the guidance of

**Dr. B.H Chandrashekar
Associate Professor
Department of MCA,
RVCE**

in partial fulfilment for the award of degree of

Master of Computer Applications

2025-2026



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DEPARTMENT OF MASTER OF COMPUTER APPLICATIONS

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CERTIFICATE

Certified that the Technical Seminar titled '**Self-Reflective Agentic AI: Enhancing Autonomous Decision-Making through Internal Self-Evaluation**' is carried out by **Swasthik Devadiga (1RV24MC108)** a bonafide student of RV College of Engineering®, Bengaluru, in partial fulfilment for the award of **Master of Computer Applications of RV College of Engineering®, Bengaluru** affiliated to **Visvesvaraya Technological University, Belagavi** during the year **2025-2026**. It is certified that all corrections/suggestions indicated for the internal assessment have been incorporated in the report deposited in the department library. The Seminar report has been approved as it satisfies the academic requirements in respect of technical seminar work prescribed by the institution for the said Degree.

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External Viva Examination

Name of Examiner

Signature with Date

1.

2.



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DECLARATION

I, **Swasthik Devadiga**, the student of Fourth Semester, Department of MCA, RV College of Engineering®, Bengaluru-560059, bearing USN: **1RV24MC108** hereby declare that the technical seminar titled '**Self-Reflective Agentic AI: Enhancing Autonomous Decision-Making through Internal Self-Evaluation**' has been carried out by me. It has been submitted in partial fulfilment of the program requirements for the award of Degree in **Master of Computer Applications** of **RV College of Engineering®, Bengaluru** affiliated to **Visvesvaraya Technological University, Belagavi** during the year **2025-2026**.

Further, I declare that the content of the seminar has not been submitted to any other university or institution for the award of any other degree or diploma.

Place: Bengaluru

Date of Submission: 28-05-2026

Signature of the Student

Student Name: Swasthik Devadiga

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ACKNOWLEDGEMENT

I express my sincere gratitude to RV College of Engineering®, Bengaluru, and the Department of Master of Computer Applications for providing me the opportunity to carry out this Technical Seminar successfully.

I would like to express my heartfelt gratitude to my internal guide, Dr. B.H Chandrashekar, Associate Professor, Department of MCA, RVCE, for the valuable guidance, continuous encouragement, constructive suggestions, and technical support provided throughout the seminar work. His guidance and insightful feedback greatly helped me in understanding the concepts and technical aspects related to the seminar topic.

I extend my sincere thanks to Dr. Jasmine KS, Associate Professor and Director, Department of MCA, RVCE, for the encouragement, motivation, and support provided during the completion of this seminar work.

I also thank all the faculty members and staff of the Department of MCA for their valuable suggestions and support throughout the seminar. Finally, I express my heartfelt gratitude to my parents, family members, and friends for their constant encouragement, motivation, and support during the successful completion of this seminar report.

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ABSTRACT

Self-Reflective Agentic Artificial Intelligence (AI) will be studied as an emerging paradigm aimed at improving the reliability, transparency, and safety of autonomous intelligent systems. With rapid advancements in Large Language Models (LLMs), autonomous agents, and reasoning-based architectures, AI systems are increasingly capable of perceiving environments, reasoning, planning, and executing actions independently. However, existing autonomous AI systems often lack internal validation mechanisms, leading to hallucinations, inconsistent reasoning, unsafe decisions, and unreliable autonomous actions. This seminar will explore the concept of integrating self-reflection into agentic AI systems, emphasizing its importance in critical applications such as intelligent assistants, robotics, healthcare systems, cybersecurity, and autonomous decision-support systems. The topic is highly relevant in current technological trends, where trustworthiness, explainability, and safe AI deployment are becoming essential requirements.

Existing AI systems and agent frameworks primarily rely on single-pass decision-making, where outputs are generated and executed without systematic internal evaluation. Techniques such as reinforcement learning, Chain-of-Thought reasoning, Self-Consistency, and external feedback loops have been proposed to improve reasoning quality and decision-making performance, yet they often identify errors only after execution. Recent research highlights the need for self-evaluating mechanisms including reasoning verification, critic-based evaluation, confidence estimation, and uncertainty analysis. However, limitations such as lack of consistency, absence of proactive error prevention, hallucinations, and dependency on human intervention remain unresolved. This seminar will analyze current models, architectures, and approaches related to autonomous reasoning systems, reflection mechanisms, Retrieval-Augmented Generation (RAG), and Explainable Artificial Intelligence (XAI), identifying gaps that motivate the adoption of self-reflective mechanisms within autonomous agents.

The seminar will focus on studying a Self-Reflective Agentic AI framework that incorporates a structured self-evaluation loop consisting of decision generation, reasoning explanation, internal critique, scoring, threshold validation, and iterative refinement. A literature survey and comparative analysis of existing techniques will be conducted to understand how self-reflection can enhance reasoning quality, transparency, safety, and autonomous decision-making reliability. The expected outcome of the study will include insights into improving decision consistency, reducing autonomous errors, enhancing trustworthiness, and enabling self-correction without external supervision. Additionally, the seminar will highlight future research directions in building trustworthy, explainable, adaptive, and human-centered AI systems, along with the broader implications of integrating reflection as a core component in next-generation intelligent autonomous agents.

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